One Service, Many Computers

Distributed systems: cooperation across machines

SCAN / LEARN

THE MAIN IDEA

A distributed system is a group of computers that cooperate over a network to provide one service. Coordination is useful, but messages can be late or lost.

FOLLOW THE SYSTEM

01

ONE REQUEST

A person asks for a page or a result.

02

MANY MACHINES

Different computers run parts of the service.

03

MESSAGES

Computers exchange data through the network.

04

ONE RESPONSE

The service combines work and replies.

USER → SERVICE → COMPUTERS → USER

COMMON MISCONCEPTION

More computers always means a faster answer.

MORE ACCURATE

Networks, shared data and coordination can slow the whole service.

THREE QUESTIONS TO KEEP

PROBLEM: How can several computers serve one application?

FAILURE: One delayed part may delay the whole answer.

IN REAL LIFE: Online shops, search, banking and AI.

SOLO DISCOVERY

Draw two servers: what happens if one loses the network?